

Homework 2 (60 points)

Due: 29/11, Fri, 11:59pm.

December 3, 2024

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Let $(\Omega, \mathcal{F}, \mathbb{P})$ is the probability space, and $(\Omega, \mathcal{F}, \mu)$ is a measurable space. Questions are testing: Modes of convergence, LLN, CLT and discrete martingale applications. But you may still need Borel-Cantelli's and convergence theorems and inequalities to finish the questions.

1. If a sequence of L^p integrable functions $X_n \rightarrow X$ a.s., the random variable X is also L^p integrable with $p \geq 1$. Moreover, if $\lim_{n \rightarrow \infty} \|X_n\|_p = \|X\|_p < \infty$ holds true, then then $X_n \rightarrow X$ in L^p , i.e. $\lim_n \|X_n - X\|_p = 0$.

Proof. we consider

$$\begin{aligned} |X_n - X|^p &= \left| X_n - \hat{X}_n + \hat{X}_n - X \right|^p \\ &\leq 2^{\frac{p}{2}} \left(\left| X_n - \hat{X}_n \right|^p + \left| \hat{X}_n - X \right|^p \right), \end{aligned}$$

where

$$\hat{X}_n = X_n 1_{|X_n| < n} \quad \text{then} \quad \hat{X}_n = X_n \text{ a.s.}$$

so we only need to find some $\hat{X}_n$ s.t. $\hat{X}_n \rightarrow X$ a.s..

By the fact that $\mathbb{E} |X_n|^p < \infty$, then we can study

$$\int \left| \hat{X}_n - X \right|^p = \int |X_n|^p 1_{|X_n| \geq n} = \int |X_n|^p - \int |X_n|^p 1_{|X_n| < n}$$

Then by MCT, we have

$$\lim_{n \rightarrow \infty} \int |X_n|^p 1_{|X_n| < n} = \int \lim_{n \rightarrow \infty} |X_n|^p 1_{|X_n| < n} = \int |X|^p.$$

So by $\lim_{n \rightarrow \infty} \|X_n\|_p = \|X\|_p < \infty$, we get the desired result. The proof is completed. $\square$

2. If $X_1, X_2, \dots$ are i.i.d. $\text{Exp}(1)$, then

$$\limsup_n \frac{X_n}{\log n} = 1, \text{ a.s.}$$

And also,

$$\lim_n \frac{1}{\log n} \left(\max_{1 \leq k \leq n} X_k \right) = 1, \text{ a.s..}$$

* Durrett, 2.3.15.

Proof. (i) Now $\mathbb{P}(X_w > x) = e^{-x}$ for all $x > 0$ and $n \geq 1$. Thus with $\varepsilon > 0$ we have

$$\mathbb{P}(X_n/\log n \geq (1 \pm \varepsilon)) = \mathbb{P}(X_n > (1 \pm \varepsilon) \log n) = n^{-(1 \pm \varepsilon)}$$

It follows that

$$\sum_{n=1}^{\infty} \mathbb{P}(X_n/\log n \geq 1 \pm \varepsilon) = \sum_{n=1}^{\infty} n^{-(1 \pm \varepsilon)} \begin{cases} < \infty & \text{for } 1 + \varepsilon \\ = \infty & \text{for } 1 - \varepsilon \end{cases}$$

Thus by the first Borel-Cantelli lemma we have

$$\limsup_n (X_n/\log n) \leq 1 \text{ a.s.}$$

and by the second Borel-Cantelli lemma $\limsup_n (X_n/\log n) \geq 1$ a.s. By combining these two results we see that $\limsup (X_m/\log n) = 1$ almost surely.

(ii) First note that since $X_n \leq \max_{1 \leq k \leq n} X_k$ we have

$$1 = \limsup_n \frac{X_n}{\log n} \leq \limsup_n \frac{\max_{1 \leq k \leq n} X_k}{\log n} \text{ a.s.} \quad (0.0.1)$$

To show that

$$\limsup_n \frac{\max_{1 \leq k \leq n} X_k}{\log n} \leq 1 \quad (0.0.2)$$

a.s., let $\varepsilon > 0$. Since $\limsup_n (X_n/\log n) \leq 1$ a.s., for all ω in a set with probability 1 we have $X_k/\log k \leq 1 + \varepsilon$ for all $k \geq$ some N_ω . However,

$$\begin{aligned} \frac{\max_{1 \leq k \leq n} X_k}{\log n} &= \max_{1 \leq k \leq n} \frac{X_k}{\log n} \\ &\leq \max \left(\max_{1 \leq k \leq N_\omega} \frac{X_k}{\log n}, \max_{N_\omega \leq k \leq n} \frac{X_k}{\log k} \right) \\ &\leq \max \left(\frac{(1 + \varepsilon) \log N_\omega}{\log n}, (1 + \varepsilon) \right) \\ &\leq \varepsilon + (1 + \varepsilon) = 1 + 2\varepsilon. \end{aligned}$$

for $n \geq \tilde{N}_\omega$ where $\tilde{N}_\omega$ is so large that $(1 + \varepsilon) \log N_\omega / \sqrt{2 \log n} < \varepsilon$ for all $n \geq \tilde{N}_\omega$. Since $\varepsilon > 0$ is arbitrary, this shows that (0.0.2) holds. Combining (0.0.1) with (0.0.2) gives the claimed limit for $\limsup_n (\max_{1 \leq k \leq n} X_k / \log n)$.

To show that $\max_{1 \leq k \leq n} X_k / \log n \rightarrow 1$ a.s., it now suffices to show that

$$\liminf_n \max_{1 \leq k \leq n} X_k / \log n \geq 1 \quad \text{a.s.}$$

equivalently, we can show that $\mathbb{P}(\max_{1 \leq k \leq n} X_k / \log n \leq (1 - \varepsilon) \text{ i.o.}) = 0$ for every $\varepsilon \in (0, 1)$. But since the X_k 's are i.i.d. $\text{Exp}(1)$, we can further compute

$$\begin{aligned} \mathbb{P}\left(\max_{1 \leq k \leq n} X_k \leq (1 - \varepsilon) \log n\right) &= \mathbb{P}\left(\cap_{k=1}^n (X_k \leq (1 - \varepsilon) \log n)\right) \\ &= \mathbb{P}(X_1 \leq (1 - \varepsilon) \log n)^n \\ &= (1 - \exp(-(1 - \varepsilon) \log n))^n = \left(1 - \frac{1}{n^{1-\varepsilon}}\right)^n \\ &= \left(1 - \frac{n^\varepsilon}{n}\right)^n \leq e^{-n^\varepsilon} \end{aligned}$$

since $1 - y \leq e^{-y}$ so that $(1 - x/n)^n \leq (e^{-x/n})^n = e^{-x}$. Thus we find that

$$\sum_{n=1}^{\infty} \mathbb{P}\left(\max_{1 \leq k \leq n} X_k \leq (1 - \varepsilon) \log n\right) \leq \sum_{n=1}^{\infty} e^{-n^\varepsilon} < \infty$$

and hence

$$\mathbb{P}\left(\max_{1 \leq k \leq n} X_k \leq (1 - \varepsilon) \log n \text{ i.o.}\right) = 0.$$

The proof is completed. □

3. * If $X_n \rightarrow X$ and $Y_n \rightarrow Y$ both in probability, prove that their products $X_n Y_n$ converges in probability.

Proof. We can compute that

$$(|X_n(Y_n - Y)| + |Y(X_n - X)| > \varepsilon) \subset (|X_n| \cdot |Y_n - Y| > \varepsilon/2) \cup (|Y| \cdot |X_n - X| > \varepsilon/2)$$

Now, for any $\lambda > 0$,

$$(|X_n| \cdot |Y_n - Y| > \varepsilon/2) \subset (|X_n - X| > 1) \cup \{|X| > \lambda\} \cup (|Y_n - Y| > \varepsilon/2(\lambda + 1)).$$

Since if none of the three conditions on the right hold true, then $|X_n| \cdot |Y_n - Y| \leq (\lambda + 1)\varepsilon/(2(\lambda + 1)) = \varepsilon/2$. By $X_n \rightarrow_p X$, $Y_n \rightarrow_p Y$, we can get for any $\lambda > 0$, it holds

$$\limsup_{n \rightarrow \infty} \mathbb{P}(|X_n| \cdot |Y_n - Y| > \varepsilon/2) \leq \mathbb{P}(|X| > \lambda) \rightarrow 0,$$

In the same way, we can get $\mathbb{P}(|Y| \cdot |X_n - X| > \varepsilon/2)$ goes to 0 when $n \rightarrow \infty$. The proof is completed. □

** If $X_n \rightarrow_{L^2} X$ and $Y_n \rightarrow_{L^2} Y$, prove that $X_n Y_n \rightarrow_{L^1} XY$.

Can we simply extend it to the conclusion: If $X_n \rightarrow_{L^p} X$ and $Y_n \rightarrow_{L^p} Y$, then that $X_n Y_n \rightarrow_{L^1} XY$ when $p > 1$? If not, provide any extra condition (if need) to achieve the statement above.

This is the special case of the following problem.

*** If $X_n \rightarrow_{L^p} X$ and also $Y_n \rightarrow_{L^q} Y$, then for some $p, q > 1$, $X_n Y_n$ converges to XY in L^1 . Prove it and find such $p, q > 1$.

Proof. Proof. At first, we can show that

$$\|X_n Y_n - XY\|_1 \leq \|X_n Y_n - X_n Y\|_1 + \|X_n Y - XY\|_1$$

Then by using Hölder's inequality, for $\frac{1}{p} + \frac{1}{q} = 1$, we can further get

$$\|X_n (Y_n - Y)\|_1 \leq \|X_n\|_p \|Y_n - Y\|_q, \quad \|X_n Y - XY\|_1 \leq \|X_n - X\|_p \|Y\|_q$$

Hence we can immediately get

$$\|X_n Y_n - XY\|_1 \rightarrow 0, \quad n \rightarrow \infty$$

by using the fact that $X_n \rightarrow X$ in L^p and $Y_n \rightarrow Y$ in L^q . We also note that $\|X_n\|_p, \|Y\|_q$ are bounded. Finally, by using Fatou's lemma, we can show that

$$\|XY\|_1 \leq \liminf_{n \rightarrow \infty} \|X_n Y_n\|_1 \leq \liminf_{n \rightarrow \infty} \|X_n\|_p \|Y_n\|_q \rightarrow 0$$

it implies $XY \in L^1$. The proof is completed. □

Can we achieve $X_n Y_n \rightarrow XY$ in L^1 when $p = q = 1$? Either prove it or provide the counter example and provide extra condition (if need) to achieve the L^1 convergence of the product.

No, there is a counter example: $X_n = Y_n = \frac{1}{w^{1/2+1/n}}$ for $w \in [0, 1]$. As $\frac{1}{w} \rightarrow \infty$ as $w \rightarrow 0$, we must use an improper integral

$$\int_0^1 \frac{1}{w} dw = \lim_{M \rightarrow 0} \int_M^1 \frac{1}{w} dw = \lim_{M \rightarrow 0} (\ln(1) - \ln(M)) = \lim_{M \rightarrow 0} -\ln(M) = \infty.$$

Thus, the integral diverges to infinity.

4. Prove that the CLT implies the WLLN directly. (We know WLLN is provided by convergence in probability, and convergence in probability implies convergence in distribution which is CLT, but how to prove the statement directly?)

Proof. Note first that $\mu_{S_n} = n\mu$, $\text{Var}(S_n) = n\sigma^2$, and so $\sigma_{S_n} = \sigma\sqrt{n}$. Now,

$$\begin{aligned} \left| \frac{S_n}{n} - \mu \right| > \varepsilon & \quad \text{if and only if} \quad |S_n - n\mu| > n\varepsilon \\ & \quad \text{if and only if} \quad \left| \frac{S_n - n\mu}{\sigma_{S_n}} \right| > \frac{n\varepsilon}{\sigma_{S_n}} \\ & \quad \text{if and only if} \quad |S_n^*| > \frac{\sqrt{n}\varepsilon}{\sigma}. \end{aligned}$$

But for any real number $\beta > 0$,

$$\frac{\sqrt{n}\varepsilon}{\sigma} > \beta$$

will hold for all large n . Hence, for any $\beta > 0$ and all large n ,

$$\mathbb{P} \left(\left| \frac{S_n}{n} - \mu \right| > \varepsilon \right) = \mathbb{P} \left(|S_n^*| > \frac{\sqrt{n}\varepsilon}{\sigma} \right) \leq \mathbb{P}(|S_n^*| > \beta)$$

So

$$\begin{aligned} \lim_{n \rightarrow \infty} \mathbb{P} \left(\left| \frac{S_n}{n} - \mu \right| > \varepsilon \right) & \leq \lim_{n \rightarrow \infty} \mathbb{P}(|S_n^*| > \beta) \\ & = \lim_{n \rightarrow \infty} \mathbb{P}(S_n^* > \beta) + \mathbb{P}(S_n^* < -\beta) \\ & = 1 - F(\beta) + F(-\beta) \end{aligned}$$

for all real numbers $\beta > 0$. By choosing β large enough, we can ensure that the CDF $F(\beta)$ is arbitrarily close to 1 and $F(-\beta)$ is arbitrarily close to 0. Hence,

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(\left| \frac{S_n}{n} - \mu \right| > \varepsilon \right) = 0$$

which is the WLLN. We completed the proof. $\square$

5. Let $X_1, X_2, \dots$ be i.i.d. with $\mathbb{E}(|X_1|) = \infty$ and let $S_n = X_1 + \dots + X_n$. Let a_n be a sequence of positive numbers with $\frac{a_n}{n}$ increasing. Then

$$\limsup_{n \rightarrow \infty} \frac{|S_n|}{a_n} = 0 \text{ or } \infty \text{ if } \sum_n \mathbb{P}(|X_1| \geq a_n) < \infty \text{ or } = \infty, \text{ respectively.}$$

* Durrett, Theorem 2.5.13.

Proof. First suppose $\sum_n \mathbb{P}(|X_1| \geq a_n) = \infty$. Since $\frac{a_n}{n}$ is increasing, we have

$$\sum_{n=1}^{\infty} \mathbb{P}(|X_1| \geq ka_n) \geq \sum_{n=1}^{\infty} \mathbb{P}(|X_1| \geq a_{kn}) \geq \frac{1}{k} \sum_{m=k}^{\infty} \mathbb{P}(|X_1| \geq a_m) = \infty$$

In particular, $\limsup_{n \rightarrow \infty} \frac{|X_n|}{a_n} = \infty$ with probability 1 by Borel-Cantelli. Since $\max(|S_{n-1}|, |S_n|) \geq \frac{|X_n|}{2}$, the claim follows.

Now suppose $\sum_n \mathbb{P}(|X_1| \geq a_n) < \infty$ - Define $Y_n = X_n 1(|X_n| < a_n)$. Since $X_n \neq Y_n$ finitely often a.s., the proof consists in checking that $\sum_n \text{Var}\left(\frac{Y_n}{a_n}\right) < \infty$ and $\frac{1}{a_n} \sum_{i=1}^d \mathbb{E}[Y_i] \rightarrow 0$.

We compute that

$$\sum_{n=1}^{\infty} \text{Var}\left(\frac{Y_n}{a_n}\right) \leq \sum_{n=1}^{\infty} \frac{\mathbb{E}[Y_n^2]}{a_n^2} = \sum_{m=1}^{\infty} \int_{[a_{m-1}, a_m)} y^2 dF(y) \sum_{n=m}^{\infty} a_n^{-2}$$

Then since $a_n \geq \frac{na_m}{m}$, $\sum_{n=m}^{\infty} a_n^{-2} \leq \frac{m^2}{a_m^2} \sum_{n=m}^{\infty} n^{-2} \leq (\text{sufficiently}) \frac{m}{a_m^2}$. Thus

$$\sum_{m=1}^{\infty} \text{Var}\left(\frac{Y_n}{a_n}\right) \leq (\text{sufficiently}) \sum_{m=1}^{\infty} m \mathbb{P}(a_{m-1} \leq |X_1| < a_m) = \sum_{m=1}^{\infty} \mathbb{P}(|X_1| > a_{m-1}) < \infty$$

To prove the mean condition, first, since $\mathbb{E}[|X_i|] = \infty$ and $\sum_{n=1}^{\infty} \mathbb{P}(|X_i| > a_n) < \infty$, we have $\frac{a_n}{n} \uparrow \infty$. Bound the following

$$\left| \frac{1}{a_n} \sum_{i=1}^d \mathbb{E}[Y_i] \right| \leq \frac{1}{a_n} \sum_{i=1}^d \mathbb{E}[|X_m| \cdot 1(|X_m| < a_m)] \leq \frac{na_N}{a_n} + \frac{n}{a_n} \mathbb{E}[|X_1| \cdot 1(a_N \leq |X_1| \leq a_n)]$$

If N grows sufficiently slowly, the first term tends to 0.

Bound $\frac{n}{a_n} \mathbb{E}[|X_1| \cdot 1(a_N \leq |X_1| \leq a_n)]$ by

$$\sum_{m=N+1}^n \frac{m}{a_m} \mathbb{E}[|X_1| \cdot 1(a_{m-1} \leq |X_1| < a_m)] \leq \sum_{m=N+1}^{\infty} m \mathbb{P}(a_{m-1} \leq |X_1| < a_m)$$

Since

$$\sum_{m=1}^{\infty} m \mathbb{P}(a_{m-1} \leq |X_1| < a_m) = \sum_{n=1}^{\infty} \mathbb{P}(|X_1| \geq a_{n-1}) < \infty$$

the latter tends to 0 as $N \rightarrow \infty$. The proof is completed. $\square$

6. We define the empirical distribution functions as the following. Let $X_1, X_2, \dots$ be i.i.d. with distribution F and let

$$F_n(x) = n^{-1} \sum_{i=1}^d 1_{(X_m \leq x)}$$

$F_n(x)$ = the observed frequency of values that are $\leq x$, hence the name given above.

Prove the Glivenko-Cantelli theorem. As $n \rightarrow \infty$,

$$\sup_x |F_n(x) - F(x)| \rightarrow 0 \quad \text{a.s.}$$

(Hence, F_n converges uniformly to F as $n \rightarrow \infty$.)

*Durrett, Theorem 2.4.9.

Proof. Fix x and let $Y_n = 1_{(X_n \leq x)}$. Since the Y_n are i.i.d. with $\mathbb{E}Y_n = \mathbb{P}(X_n \leq x) = F(x)$, the strong law of large numbers implies that $F_n(x) = n^{-1} \sum_{i=1}^n Y_i \rightarrow F(x)$ a.s. In general, if F_n is a sequence of nondecreasing functions that converges pointwise to a bounded and continuous limit F then $\sup_x |F_n(x) - F(x)| \rightarrow 0$. However, the distribution function $F(x)$ may have jumps, so we have to work a little harder.

Again, fix x and let $Z_n = 1_{(X_n < x)}$. Since the Z_n are i.i.d. with $\mathbb{E}Z_n = \mathbb{P}(X_n < x) = F(x-) = \lim_{y \uparrow x} F(y)$, the strong law of large numbers implies that $F_n(x-) = n^{-1} \sum_{i=1}^n Z_i \rightarrow F(x-)$ a.s. For $1 \leq j \leq k-1$ let $x_{j,k} = \inf\{y : F(y) \geq j/k\}$. The pointwise convergence of $F_n(x)$ and $F_n(x-)$ imply that we can pick $N_k(\omega)$ so that if $n \geq N_k(\omega)$ then

$$|F_n(x_{j,k}) - F(x_{j,k})| < k^{-1} \quad \text{and} \quad |F_n(x_{j,k}-) - F(x_{j,k}-)| < k^{-1}$$

for $1 \leq j \leq k-1$. If we let $x_{0,k} = -\infty$ and $x_{k,k} = \infty$, then the last two inequalities hold for $j=0$ or k . If $x \in (x_{j-1,k}, x_{j,k})$ with $1 \leq j \leq k$ and $n \geq N_k(\omega)$, then using the monotonicity of F_n and F , and $F(x_{j,k}-) - F(x_{j-1,k}) \leq k^{-1}$, we have

$$\begin{aligned} F_n(x) &\leq F_n(x_{j,k}-) \leq F(x_{j,k}-) + k^{-1} \leq F(x_{j-1,k}) + 2k^{-1} \leq F(x) + 2k^{-1} \\ F_n(x) &\geq F_n(x_{j-1,k}) \geq F(x_{j-1,k}) - k^{-1} \geq F(x_{j,k}-) - 2k^{-1} \geq F(x) - 2k^{-1} \end{aligned}$$

so $\sup_x |F_n(x) - F(x)| \leq 2k^{-1}$, and we have proved the result. $\square$

7. Show that if $X = (X^1, \dots, X^d)$ is a single d -dimensional vector such that uniformly distributed over the surface of the sphere of radius $\sqrt{d}$ in $\mathbb{R}^d$ then $X^i \sim \mathcal{N}(0, 1)$.

Hint: (1). If $Z \sim_{i.i.d.} \mathcal{N}(0, I_d)$, then we can have $f(Z) = \left(\frac{1}{\sqrt{2\pi}}\right)^{d/2} e^{-\frac{\|Z\|^2}{2}}$, where $\|Z\|^2 = Z^T Z$.

(2). For any orthogonal matrix O , we can have $\|Ox\|^2 = \|x\|^2$.

(3). We also note that uniformly distributed over the surface of the unit sphere is the ONLY distribution which is invariant to rotation (so preserve the distribution) and belongs to UNIT sphere.

(4). Let $Y_1, Y_2, \dots$ be i.i.d. standard normals and let $X^i = Y_i \left(d / \sum_{i=1}^d Y_i^2\right)^{1/2}$.

* Durrett, Theorem 3.2.15.

Proof. Let $X^i = Y_i / \left(\sum_{i=1}^d Y_i^2 / d \right)^{1/2}$, where $Y_i \sim_{i.i.d.} \mathcal{N}(0, 1)$ then the random vector is $Y = (Y_1, \dots, Y_n) \sim \mathcal{N}(0, I_d)$.

Since the uniformly distributed over the surface of the unit sphere is the only distribution which is invariant to rotation and belongs to unit sphere. For any orthogonal matrix A , we have $\frac{Y}{\|Y\|_2} \stackrel{\mathcal{D}}{=} \frac{AY}{\|AY\|_2}$. Thus,

$$\frac{Y}{\|Y\|_2} = \left(Y_1 / \sum_{i=1}^d Y_i^2, \dots, Y_i / \sum_{i=1}^d Y_i^2 \right)$$

is uniformly distributed over the surface of the unit sphere. Then, $X_n \stackrel{\mathcal{D}}{=} \sqrt{d} \frac{Y}{\|Y\|_2}$, and thus we can get

$$X^i \stackrel{\mathcal{D}}{=} Y_i / \left(\sum_{i=1}^d Y_i^2 / d \right)^{1/2}.$$

So we can get $Y_i / \left(\sum_{i=1}^d Y_i^2 / n \right)^{1/2} \rightarrow_{\text{weak}} \mathcal{N}(0, 1)$. For $Y_i \sim \mathcal{N}(0, 1)$, $\mathbb{E}Y_i^2 = 1 < \infty$. Thus, by WLLN,

$$\sum_{i=1}^d Y_i^2 / d \rightarrow_p 1.$$

weo we get $\sum_{i=1}^d Y_i^2 / n \rightarrow_{\text{weak}} 1$ by the modes of convergence. Note that $f(x) = \frac{1}{\sqrt{x}} 1_{(x>0)}$ is continuous on $\mathbb{R} \setminus \{0\}$ and $\mathbb{P} \left(\sum_{i=1}^d Y_i^2 / d = 0 \right) = \mathbb{P}(Y_1 = \dots = Y_d = 0) = 0$. By Continuous Mapping Theorem, we get

$$\left(\sum_{i=1}^d Y_i^2 / n \right)^{-1/2} \rightarrow_{\text{weak}} 1,$$

hence, we get

$$Y_i \cdot \left(\sum_{i=1}^d Y_i^2 / n \right)^{-1/2} \rightarrow_{\text{weak}} \mathcal{N}(0, 1)$$

Thus, $X^i \rightarrow_{\text{weak}} \mathcal{N}(0, 1)$. The proof is completed. $\square$

8. Recall we study WLLN by assuming the sequence $\{X_n\}_{n=1}^\infty$ is i.i.d., the question is that if it is possible to drop the condition on i.i.d.. Suppose the mean $\mu = \mathbb{E}[X_n]$, we can define a function to measure the pairwise dependence X_i and X_j in the sense that

$$\varphi(i, j) = \mathbb{E}[(X_i - \mu)(X_j - \mu)].$$

* Suppose the sum $\max_i \sum_j |\varphi(i, j)| < \infty$, prove the WLLN.

Proof. Let S_T be the sum of the first T Y_t s and set $\bar{Y}_T := S_T/T$. The variance of S_T is

$$\begin{aligned} \mathbb{E} \left[(S_T - T\mu)^2 \right] &= \sum_{s=0}^{T-1} \sum_{t=0}^{T-1} \mathbb{E} [(Y_s - \mu)(Y_t - \mu)] \\ &\leq \sum_{s=0}^{T-1} \sum_{t=-\infty}^{\infty} |\varphi(s, t)| \leq Tc. \end{aligned}$$

so $\bar{Y}_T$ has variance $\text{Var} [\bar{Y}_T] \leq c/T$ and by Chebychev's inequality

$$\begin{aligned} \mathbb{P} [|\bar{Y}_T - \mu| > \varepsilon] &\leq \frac{\mathbb{E} [(\bar{Y}_T - \mu)^2]}{\varepsilon^2} = \frac{\mathbb{E} [(S_T - T\mu)^2]}{T^2 \varepsilon^2} \\ &\leq \frac{Tc}{T^2 \varepsilon^2} = \frac{c}{T \varepsilon^2} \rightarrow 0 \quad \text{as } T \rightarrow \infty. \end{aligned}$$

The proof is completed. $\square$

** Suppose $\varphi(i, j)$ perform in a sort of decay trend to 0 such as $\varphi(i, j) = r^{|i-j|}$ for $0 < r < 1$, prove the WLLN.

Proof. We can rewrite the sum as follows:

$$\sum_j r^{|i-j|} = \sum_{j < i} r^{i-j} + \sum_{j > i} r^{j-i} + r^{|i-i|} = \sum_{k=0}^{\infty} r^k + \sum_{k=1}^{\infty} r^k$$

where $k = i - j$ for the first sum (when $j < i$) and $k = j - i$ for the second sum (when $j > i$).

For $j < i$ (or $k = i - j$), we have

$$\sum_{k=0}^{\infty} r^k = \frac{1}{1-r} \quad (\text{geometric series})$$

For $j > i$ (or $k = j - i$), we also have

$$\sum_{k=1}^{\infty} r^k = r \sum_{k=0}^{\infty} r^k = \frac{r}{1-r} \quad (\text{geometric series})$$

So we can compute that

$$\sum_j r^{|i-j|} = \sum_{j < i} r^{|i-j|} + \sum_{j > i} r^{|i-j|} + 1 = \frac{1}{1-r} + \frac{r}{1-r} + 1$$

it gives

$$= \frac{1}{1-r} + \frac{r}{1-r} + 1 = \frac{1+r}{1-r} + 1 = \frac{1+r+1-r}{1-r} = \frac{2}{1-r},$$

which does not depend on i . This shows that the maximum is indeed a finite value for $0 < r < 1$ and thus is convergent. So by the first problem, we can get the proof done. $\square$

Remark: We may have other convergent forms to describe $\varphi(i, j)$ term.

Remark for your interests: Why in high dimension world, the independence becomes not so bothering? By uniformly sampling, we get a random uniform vector, from question 7, we have $X^d = \sqrt{d} \frac{Z}{\|Z\|} = \frac{Z}{\|Z\|/\sqrt{d}}$, a larger d implies $X \rightarrow_{\text{weak}} Z$.

9. Let $X_1, X_2, \dots$ be i.i.d. with $\mathbb{E}X_i = 0$ and $\mathbb{E}X_i^2 = \sigma^2 \in (0, \infty)$, and let $S_n = X_1 + \dots + X_n$. Let N_n be a sequence of nonnegative integer-valued random variables and a_n a sequence of integers with $a_n \rightarrow \infty$ and $N_n/a_n \rightarrow 1$ in probability. Show that

$$\frac{S_{N_n}}{\sigma\sqrt{a_n}} \rightarrow_{\text{weak}} Z, \quad Z \sim \mathcal{N}(0, 1).$$

Hint: Use Kolmogorov's maximal inequality to conclude that for $Y_n = \frac{S_{N_n}}{\sigma\sqrt{a_n}}$ and $Z_n = \frac{S_{a_n}}{\sigma\sqrt{a_n}}$, we have $Y_n - Z_n \rightarrow 0$ in probability.

* Durrett, Ex.3.4.6.

Proof. By CLT for i.i.d. sequence, we know that $\frac{S_n}{\sigma\sqrt{n}} \rightarrow_{\text{weak}} \mathcal{N}(0, 1)$. Thus, its subsequence $\frac{S_{a_n}}{\sigma\sqrt{a_n}} \rightarrow_{\text{weak}} \mathcal{N}(0, 1)$. For Y_n and Z_n defined above, if one can show that $Y_n - Z_n \rightarrow_p 0$, then we can get

$$\frac{S_{N_n}}{\sigma\sqrt{a_n}} = \frac{S_{a_n}}{\sigma\sqrt{a_n}} + \frac{S_{a_n} - S_{N_n}}{\sigma\sqrt{a_n}} = \frac{S_{a_n}}{\sigma\sqrt{a_n}} + (Y_n - Z_n) \rightarrow_{\text{weak}} \mathcal{N}(0, 1).$$

We show the claim: $Y_n - Z_n \rightarrow_p 0$.

Fix $\varepsilon > 0$. Since $\frac{N_n}{a_n} \rightarrow_p 1$, $\mathbb{P}\left(\left|\frac{N_n}{a_n} - 1\right| > \varepsilon\right) = \mathbb{P}(|N_n - a_n| > a_n\varepsilon) \rightarrow 0$. Then,

$$\begin{aligned} & \limsup_n \mathbb{P}(|Y_n - Z_n| > \delta) \\ &= \limsup_n [\mathbb{P}(|Y_n - Z_n| > \delta, |N_n - a_n| > a_n\varepsilon) + \mathbb{P}(|Y_n - Z_n| > \delta, |N_n - a_n| \leq a_n\varepsilon)] \\ &\leq \limsup_n \mathbb{P}(|Y_n - Z_n| > \delta, |N_n - a_n| > a_n\varepsilon) \\ &\quad + \limsup_n \mathbb{P}(|Y_n - Z_n| > \delta, |N_n - a_n| \leq a_n\varepsilon) \\ &= \limsup_n \mathbb{P}(|Y_n - Z_n| > \delta, |N_n - a_n| \leq a_n\varepsilon). \end{aligned}$$

Note that

$$\mathbb{P}(|Y_n - Z_n| > \delta, |N_n - a_n| \leq a_n \varepsilon) = \mathbb{P}(|S_{N_n} - S_{a_n}| > \delta \cdot \sigma \sqrt{a_n}, |N_n - a_n| \leq a_n \varepsilon).$$

Since $|N_n - a_n| \leq a_n \varepsilon$,

$$\begin{aligned} & \mathbb{P}(|S_{N_n} - S_{a_n}| > \delta \sigma \sqrt{a_n}, |N_n - a_n| \leq a_n \varepsilon) \\ & \leq \mathbb{P}\left(\max_{(1-\varepsilon)a_n \leq m \leq (1+\varepsilon)a_n} |S_m - S_{[(1-\varepsilon)a_n]}| > \frac{\delta \sigma \sqrt{a_n}}{2}\right) \\ & \leq \frac{4}{\delta^2 \sigma^2 a_n} \text{Var}(S_{[(1+\varepsilon)a_n]} - S_{[(1-\varepsilon)a_n]}) \\ & = \frac{4}{\delta^2 \sigma^2 a_n} \sigma^2 ([(1+\varepsilon)a_n] - [(1-\varepsilon)a_n]) \\ & \leq \frac{8\varepsilon}{\delta^2} \end{aligned}$$

The second inequality holds by Kolmogorov's Maximal Inequality. Thus,

$$\limsup_n \mathbb{P}(|Y_n - Z_n| > \delta) \leq \frac{8\varepsilon}{\delta^2}.$$

Since ε was arbitrarily chosen, $\mathbb{P}(|Y_n - Z_n| > \delta) \rightarrow 0$, for any $\delta > 0$. Therefore, $Y_n \rightarrow_p Z_n$. The proof is good to go. $\square$

10. Let $(Y_n \mid n \geq 1)$ be a sequence of independent random variables with mean 0 and variance σ^2 . Show that $\left(X_n := \left(\sum_{j=1}^n Y_j\right)^2 - n\sigma^2 \mid n \geq 1\right)$ is a martingale.

Proof. To show that $(X_n := \left(\sum_{j=1}^n Y_j\right)^2 - n\sigma^2)$ is a martingale, we need to verify the martingale property:

$$\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = X_n$$

where $\mathcal{F}_n$ is the σ -algebra generated by $Y_1, Y_2, \dots, Y_n$. We can compute that

$$\begin{aligned} X_{n+1} &= \left(\sum_{j=1}^{n+1} Y_j\right)^2 - (n+1)\sigma^2 \\ &= \left(\sum_{j=1}^n Y_j + Y_{n+1}\right)^2 - (n+1)\sigma^2 \\ &= \left(\sum_{j=1}^n Y_j\right)^2 + 2Y_{n+1} \sum_{j=1}^n Y_j + Y_{n+1}^2 - n\sigma^2 - \sigma^2. \end{aligned}$$

Since Y_{n+1} is independent of $\mathcal{F}_n$, it implies

$$\mathbb{E}[Y_{n+1} \mid \mathcal{F}_n] = 0, \quad \mathbb{E}[Y_{n+1}^2 \mid \mathcal{F}_n] = \sigma^2.$$

and thus we have

$$\begin{aligned} \mathbb{E}[X_{n+1} \mid \mathcal{F}_n] &= \mathbb{E} \left[\left(\sum_{j=1}^n Y_j \right)^2 + 2Y_{n+1} \sum_{j=1}^n Y_j + Y_{n+1}^2 - (n+1)\sigma^2 \mid \mathcal{F}_n \right] \\ &= \left(\sum_{j=1}^n Y_j \right)^2 + 2\mathbb{E}[Y_{n+1} \mid \mathcal{F}_n] \sum_{j=1}^n Y_j + \mathbb{E}[Y_{n+1}^2 \mid \mathcal{F}_n] - (n+1)\sigma^2 \\ &= \left(\sum_{j=1}^n Y_j \right)^2 + 0 + \sigma^2 - (n+1)\sigma^2 \\ &= \left(\sum_{j=1}^n Y_j \right)^2 - n\sigma^2 \\ &= X_n, \end{aligned}$$

so the sequence (X_n) is indeed a martingale. The proof is completed. $\square$

11. Galton-Watson Branching Process. Let $\xi_{n,k}, n = 1, 2, \dots, k = 1, 2, \dots$ be i.i.d. nonnegative integer random variables with finite mean μ and variance σ^2 . Define the Galton-Watson branching process

$$Z_0 := 1, \quad Z_{n+1} = \sum_{k=1}^{Z_n} \xi_{n+1,k},$$

and let $\mathcal{F}_n := \sigma(Z_j : 0 \leq j \leq n), n \geq 0$ be the natural filtration.

* Prove that $M_n := Z_n/\mu^n, n \geq 0$ is an $(\mathcal{F}_n)_{n \geq 0}$ -martingale.

Proof. We can use the linearity of expectation and the independent to compute that

$$\mathbb{E}[Z_{n+1} \mid \mathcal{F}_n] = Z_n \cdot \mathbb{E}[\xi_{1,1}] = Z_n \mu.$$

Then we can easily find

$$\mathbb{E} \left[\frac{Z_{n+1}}{\mu^{n+1}} \mid \mathcal{F}_n \right] = \frac{1}{\mu^{n+1}} \mathbb{E}[Z_{n+1} \mid \mathcal{F}_n] = \frac{Z_n \mu}{\mu^{n+1}} = \frac{Z_n}{\mu^n} = M_n.$$

The proof is straightforward. $\square$

** Prove that $\mathbb{E}(Z_{n+1}^2 \mid \mathcal{F}_n) = \mu^2 Z_n^2 + \sigma^2 Z_n$.

Proof. We first compute that

$$Z_{n+1}^2 = \left(\sum_{k=1}^{Z_n} \xi_{n+1,k} \right)^2 = \sum_{k=1}^{Z_n} \xi_{n+1,k}^2 + 2 \sum_{1 \leq i < j \leq Z_n} \xi_{n+1,i} \xi_{n+1,j},$$

hence, we have

$$\mathbb{E}[Z_{n+1}^2 \mid \mathcal{F}_n] = \sum_{k=1}^{Z_n} \mathbb{E}[\xi_{n+1,k}^2] + 2 \sum_{1 \leq i < j \leq Z_n} \mathbb{E}[\xi_{n+1,i}] \mathbb{E}[\xi_{n+1,j}].$$

Given $\mathbb{E}[\xi_{n+1,k}] = \mu$ and $\text{Var}(\xi_{n+1,k}) = \sigma^2$, we further get

$$\mathbb{E}[\xi_{n+1,k}^2] = \sigma^2 + \mu^2.$$

So we get

$$\mathbb{E}[Z_{n+1}^2 \mid \mathcal{F}_n] = Z_n(\sigma^2 + \mu^2) + 2 \binom{Z_n}{2} \mu^2 = \mu^2 Z_n^2 + \sigma^2 Z_n.$$

The proof is done. □

*** Using the result from ** prove that

$$N_n := \begin{cases} M_n^2 - \frac{\sigma^2}{\mu^{n+1}} \frac{\mu^n - 1}{\mu - 1} M_n, & \text{if } \mu \neq 1, \\ M_n^2 - n\sigma^2 M_n, & \text{if } \mu = 1 \end{cases}$$

is also an $(\mathcal{F}_n)_{n \geq 0}$ -martingale.

Proof. From part **, we have:

$$\mathbb{E}[Z_{n+1}^2 \mid \mathcal{F}_n] = \mu^2 Z_n^2 + \sigma^2 Z_n$$

Thus:

$$\mathbb{E}[M_{n+1}^2 \mid \mathcal{F}_n] = \mathbb{E} \left[\left(\frac{Z_{n+1}}{\mu^{n+1}} \right)^2 \middle| \mathcal{F}_n \right] = \frac{\mu^2 Z_n^2 + \sigma^2 Z_n}{\mu^{2(n+1)}}$$

Hence,

$$\begin{aligned} \mathbb{E}[N_{n+1} \mid \mathcal{F}_n] &= \mathbb{E}[M_{n+1}^2 \mid \mathcal{F}_n] - \frac{\sigma^2}{\mu^{n+2}} \frac{\mu^{n+1} - 1}{\mu - 1} \mathbb{E}[M_{n+1} \mid \mathcal{F}_n] \\ &= \frac{\mu^2 Z_n^2 + \sigma^2 Z_n}{\mu^{2(n+1)}} - \frac{\sigma^2}{\mu^{n+2}} \frac{\mu^{n+1} - 1}{\mu - 1} M_n \\ &= N_n. \end{aligned}$$

If $\mu = 1$, we have

$$\mathbb{E}[M_{n+1}^2 \mid \mathcal{F}_n] = \mathbb{E}[Z_{n+1}^2 \mid \mathcal{F}_n] = Z_n^2 + \sigma^2 Z_n$$

So we get

$$\mathbb{E}[N_{n+1} \mid \mathcal{F}_n] = \mathbb{E}[M_{n+1}^2 \mid \mathcal{F}_n] - (n+1)\sigma^2 \mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = Z_n^2 + \sigma^2 Z_n - (n+1)\sigma^2 M_n = N_n$$

Both cases show $\mathbb{E}[N_{n+1} \mid \mathcal{F}_n] = N_n$, so N_n is a martingale. The proof is completed. $\square$

**** Using the result from *** prove that if $\mu > 1$ then M_n is bounded in $\mathcal{L}^2$, while if $\mu \leq 1$ then $\lim_{n \rightarrow \infty} \mathbb{E}M_n^2 = \infty$.

Proof. For $\mu > 1$, we showed above that N_n is a martingale, then

$$\mathbb{E}[N_n] = \mathbb{E}[N_0] = 1, \quad \text{for all } n.$$

It implies M_n is bounded in $\mathcal{L}^2$ because N_n controls M_n^2 .

If $\mu = 1$, we have that

$$1 = \mathbb{E}[N_0] = \mathbb{E}[N_n] = \mathbb{E}[M_n^2] - n\sigma^2 \mathbb{E}[M_n] = \mathbb{E}[M_n^2] - n\sigma^2 \mathbb{E}[M_0] = \mathbb{E}[M_n^2] - n\sigma^2.$$

Now if $\mu < 1$, we also can compute that

$$\begin{aligned} N_n &= M_n^2 - \frac{\sigma^2}{\mu^{n+1}} \frac{\mu^n - 1}{\mu - 1} M_n = M_n^2 - \frac{\sigma^2}{\mu^{n+1}} (1 + \mu + \mu^2 + \dots + \mu^{n-1}) M_n \\ &\geq M_n^2 - \frac{n\sigma^2}{\mu^{n+1}} M_n, \quad \text{since } \mu < 1. \end{aligned}$$

In the same line of $\mu = 1$, we have the $\mathbb{E}[M_n^2]$ blows up. The proof is completed. $\square$

12. Gambler's ruin. Let $X_1, X_2, \dots$ be i.i.d. random variables with $\mathbb{P}(X_i = 1) = 1 - \mathbb{P}(X_i = -1) = 1/2$. Fix also $0 < b < a$ integers, and

$$S_0 = b, \quad S_n := \sum_{k=1}^n X_k, \quad T := \inf(n : S_n = 0 \text{ or } S_n = a).$$

* Show that both

$$M_n := S_n \quad \text{and} \quad N_n := S_n^2 - n$$

are martingales w.r.t. the natural filtration.

Proof. We have $S_n = \sum_{k=1}^n X_k$ is a sum of i.i.d. random variables. So we can compute that $\mathbb{E}[X_i] = 0$ since $\mathbb{P}(X_i = 1) = \mathbb{P}(X_i = -1) = \frac{1}{2}$. Therefore,

$$\mathbb{E}[S_{n+1} \mid \mathcal{F}_n] = S_n + \mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = S_n + 0 = S_n.$$

Hence, $M_n = S_n$ is a martingale. We have

$$S_{n+1}^2 = (S_n + X_{n+1})^2 = S_n^2 + 2S_n X_{n+1} + X_{n+1}^2.$$

Then,

$$\mathbb{E}[S_{n+1}^2 \mid \mathcal{F}_n] = S_n^2 + 2S_n \mathbb{E}[X_{n+1} \mid \mathcal{F}_n] + \mathbb{E}[X_{n+1}^2 \mid \mathcal{F}_n].$$

Since $\mathbb{E}[X_{n+1}] = 0$ and $\mathbb{E}[X_{n+1}^2] = 1$, it follows that $\mathbb{E}[S_{n+1}^2 \mid \mathcal{F}_n] = S_n^2 + 1$. Thus,

$$\mathbb{E}[N_{n+1} \mid \mathcal{F}_n] = \mathbb{E}[S_{n+1}^2 - (n+1) \mid \mathcal{F}_n] = S_n^2 - n = N_n.$$

Hence, $N_n = S_n^2 - n$ is a martingale. The proof is completed. $\square$

** Compute the ruin probability $\mathbb{P}(S_T = 0)$ when she stops the game.

Define the stopping times

$$T_\ell := \inf \{n > 0 : S_n = 0\}, \quad T_w := \inf \{n > 0 : S_n = a\}, \quad T := \min \{T_\ell, T_w\}.$$

Note that $\{\text{the gambler wins } a - b \text{ pounds}\} = \{T = T_w\}$,
 $\{\text{the gambler loses } b \text{ pounds}\} = \{T = T_\ell\}.$

By the Optional Stopping Theorem

$$\mathbb{E}(S_T) = \mathbb{E}(S_0) = 0.$$

Hence

$$-b\mathbb{P}(T = T_\ell) + (a - b)\mathbb{P}(T = T_w) = 0.$$

On the other hand,

$$\mathbb{P}(T = T_\ell) + \mathbb{P}(T = T_w) = 1.$$

Solving the last two equations we get

$$\mathbb{P}(T = T_\ell) = \frac{a - b}{a}, \quad \mathbb{P}(T = T_w) = \frac{b}{a}.$$

So $\mathbb{P}(S_n = 0) = \mathbb{P}(T = T_\ell) = \frac{a-b}{a}$, we get the result.

*** Compute the expected number of bets $\mathbb{E}T$ before she stops the game.

We have that $M_n := S_n^2 - n$ is a martingale also. Now, apply the Optional Stopping Theorem we have

$$b^2 = S_0^2 = \mathbb{E}(M_T) = \mathbb{E}(S_T^2 - T) = \mathbb{P}(T = T_\ell) \cdot 0 + \mathbb{P}(T = T_w) a^2 - \mathbb{E}(T).$$

Hence, using the result from ** we get

$$\mathbb{E}T = a^2 \mathbb{P}(T = T_w) = ab - b^2.$$

The problem is done.